

Customer Churn Prediction

Using Machine Learning

Prepared by: Rand alhasnani

Introduction

What is Customer Churn?

- Occurs when customers stop using a company's services
- Predicting churn helps improve customer retention
- ML can identify at-risk customers early

Project Objective

Predict whether a customer will leave the company or remain, enabling proactive retention strategies.

7,043

Customers

21

Columns

Binary

Target

Dataset Description

Source: Kaggle

Target Variable (Churn)

No — Customer stays

Yes — Customer leaves

Telco Customer Churn Dataset

Customers	**7,043**
Columns	**21**
Target Variable	**Binary (Churn)**

df.head() — Sample Data

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	...
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	...
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	...
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	...
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	...

5 rows × 21 columns

Data Preprocessing

1 Remove customerID

Non-informative unique identifier removed from features

2 Convert TotalCharges

Converted from string to numeric format for analysis

3 Handle Missing Values

11 missing values replaced using median imputation

4 One-Hot Encoding

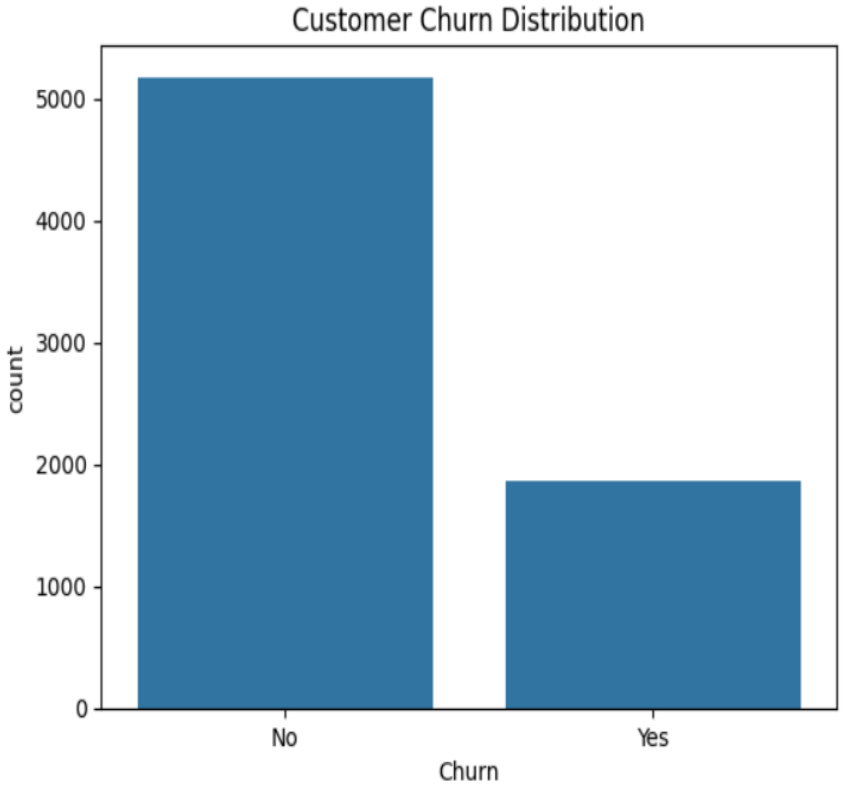
Categorical variables encoded into binary columns

5 Encode Target Variable

Churn converted to binary: Yes = 1, No = 0

✓ **Result: Clean dataset · 80/20 train-test split · features scaled for Logistic Regression**

Churn Distribution



73.46%

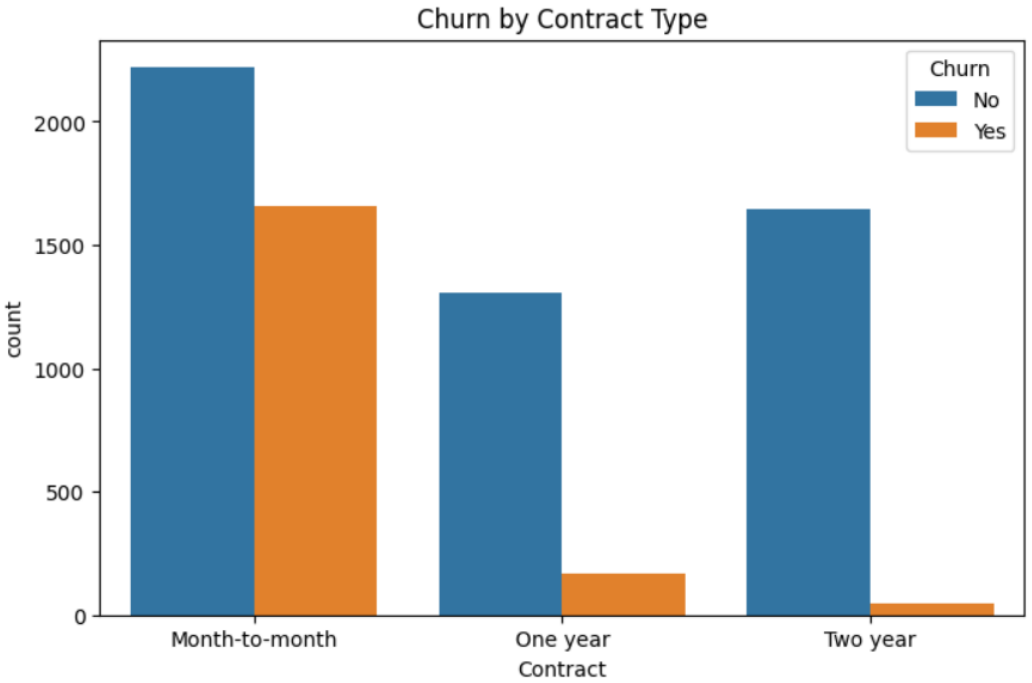
Stayed · 5,174 customers

26.54%

Churned · 1,869 customers

Insight: Class imbalance noted — important for model evaluation

Contract Type vs Churn

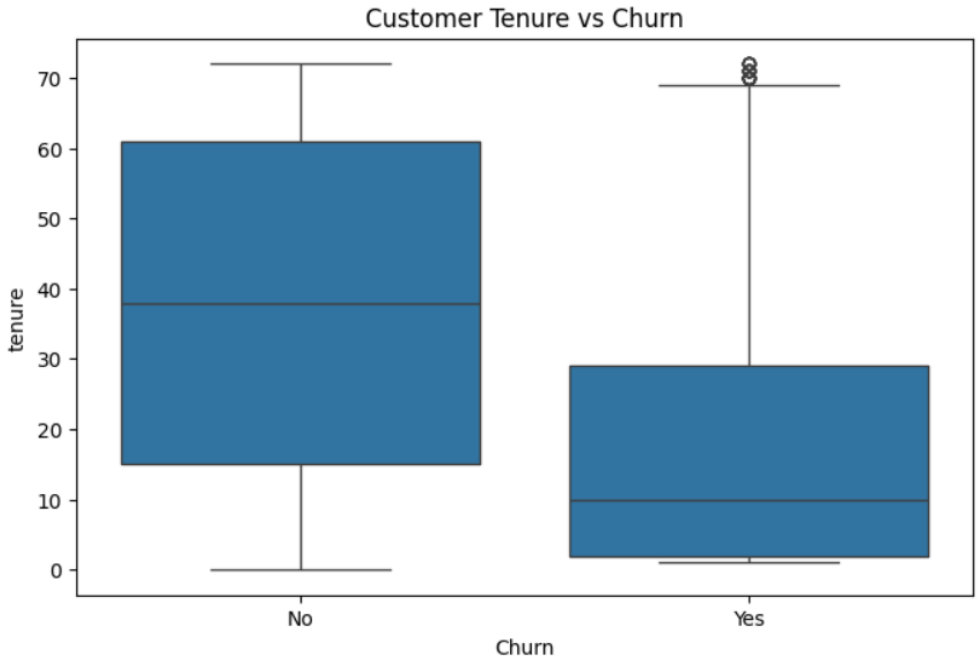


Key Findings

- Month-to-month contracts had the highest churn (~43%)
- One-year contracts showed moderate churn (~11%)
- Two-year contracts had the lowest churn (~3%)

Long-term contracts significantly increase customer retention

Tenure vs Churn



Findings

~18 mo

Avg. Tenure (Churned)

~38 mo

Avg. Tenure (Stayed)

Insight: Customer loyalty increases with subscription duration. New customers need extra attention.

Machine Learning Models

Logistic Regression

81.97%

Accuracy · ROC-AUC 0.862 · Recall 0.60

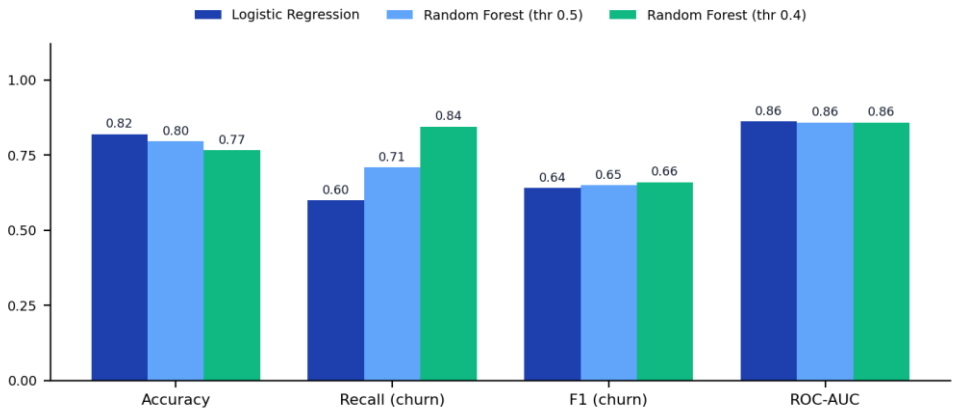
Random Forest

79.56%

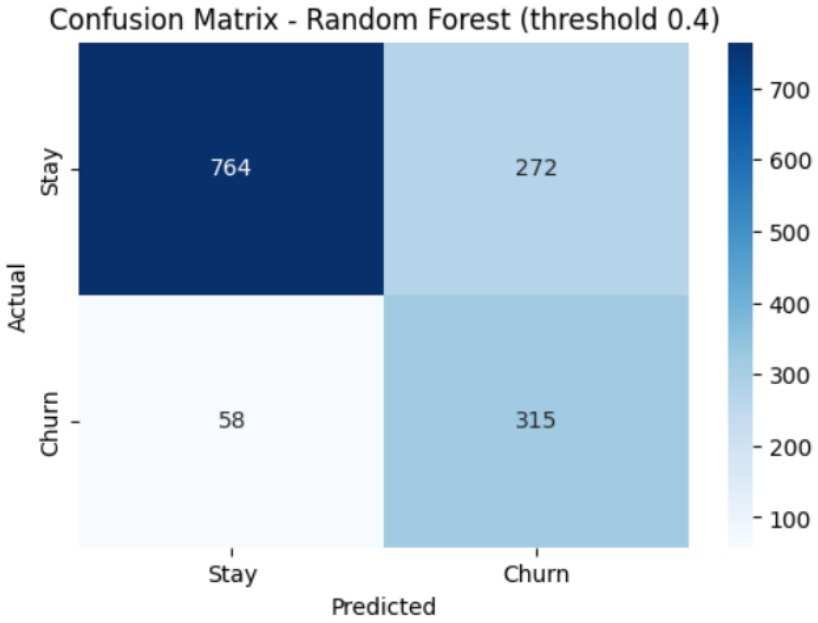
Accuracy · ROC-AUC 0.859 · Recall 0.71

Model Comparison

Model	Accuracy	Recall	ROC-AUC
Logistic Regression	81.97%	0.60	0.862
Random Forest	79.56%	0.71	0.859
RF · threshold 0.4	**76.58%**	**0.84**	**0.859**



Final Model & Threshold Tuning



Random Forest: Effect of Decision Threshold

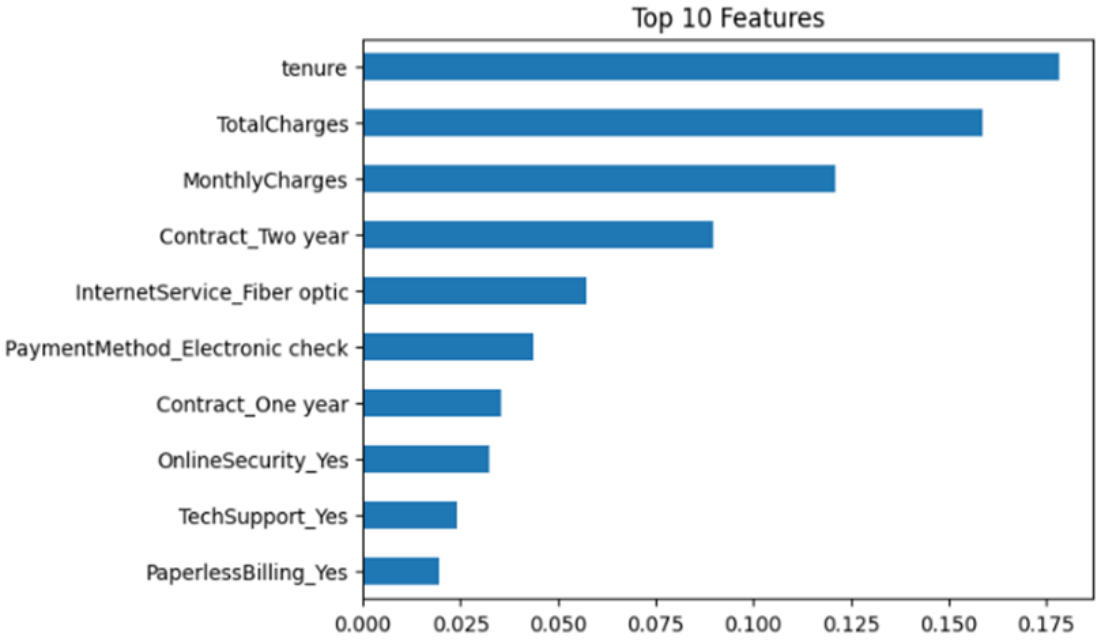
Threshold	Precision	Recall	F1
0.3	0.47	0.88	0.61
0.4 (selected)	**0.54**	**0.84**	**0.66**
0.5	0.60	0.71	0.65
0.6	0.65	0.58	0.61

Why threshold 0.4?

Lowering the threshold catches more churners: 315 of 373 are detected (recall 84%).

Trade-off: 272 loyal customers are flagged by mistake (precision 54%).

Key Churn Drivers



What drives churn?

Tenure: new customers are the most at risk.

Charges: TotalCharges and MonthlyCharges rank 2nd and 3rd.

Contract: month-to-month churns most (~43%) vs two-year (~3%).

Services: fiber optic, electronic check, online security and tech support also matter.

Conclusion

Random Forest (threshold 0.4) detects 84% of churning customers

Logistic Regression had the highest accuracy (81.97%) and ROC-AUC (0.862)

Tenure, charges and contract type were the top churn factors

Companies can apply these insights to reduce customer loss

Thank You · Questions Welcome